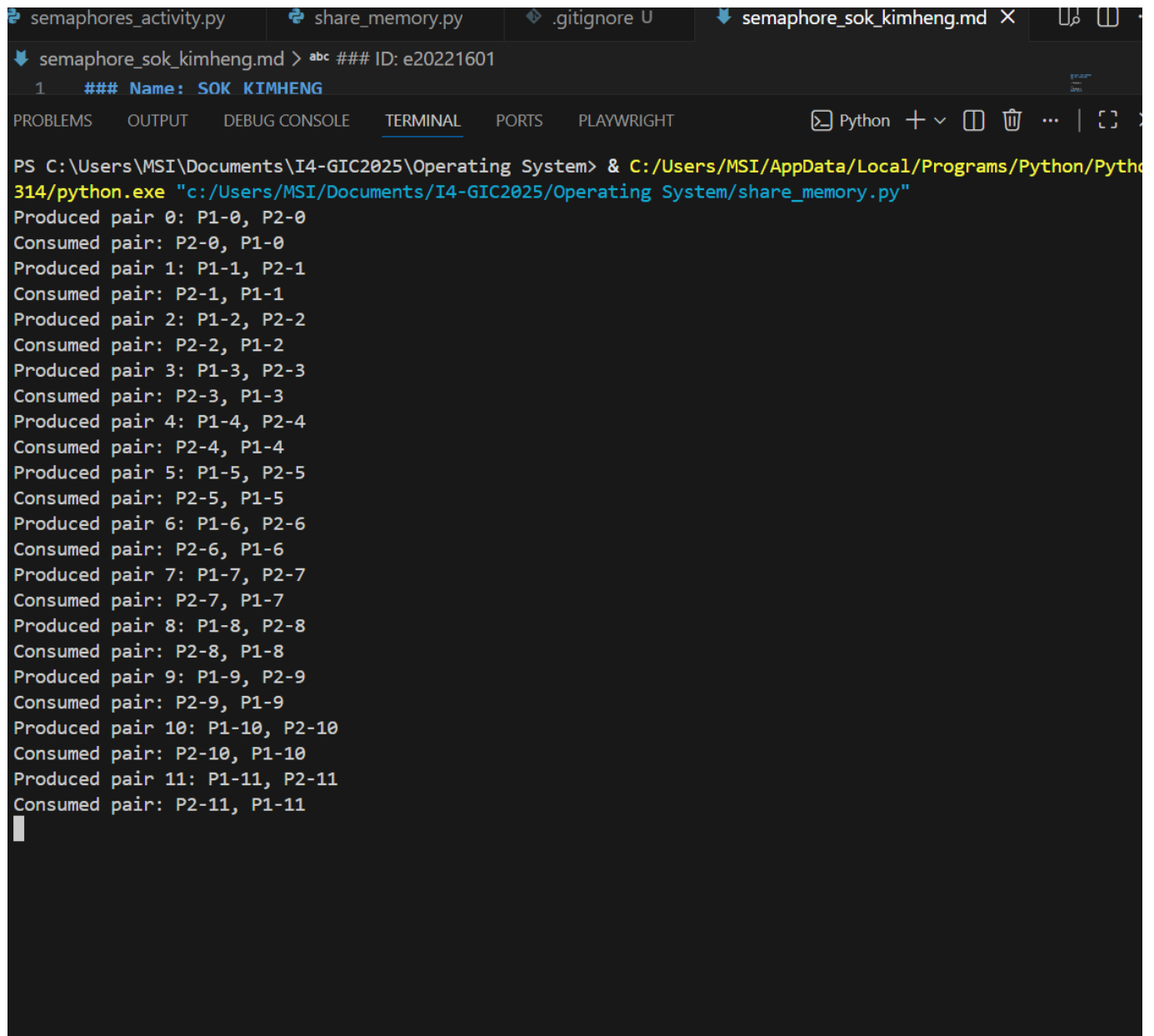


Name: SOK KIMHENG

ID: e20221601

- Problem 1

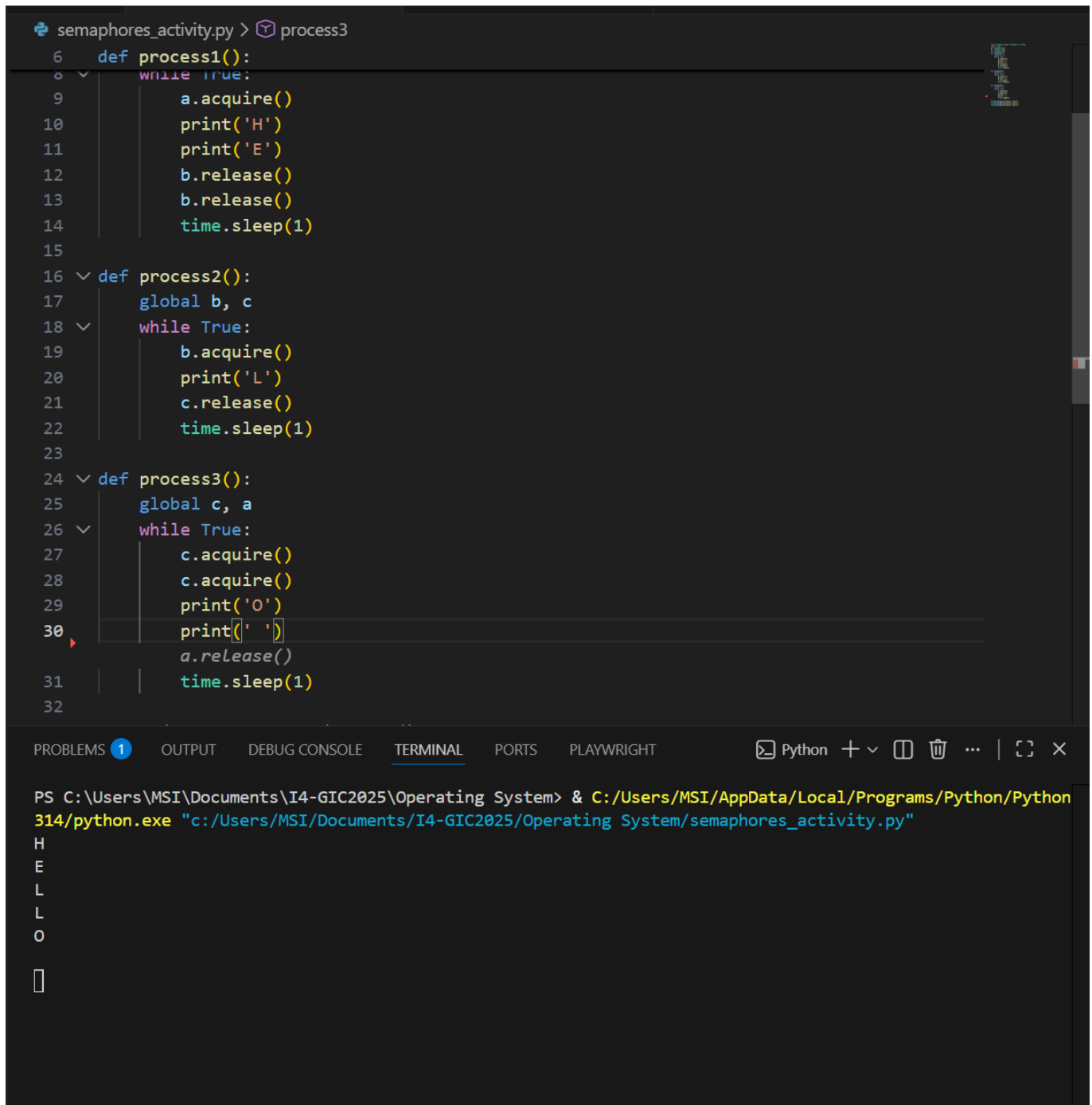
Result



```
semaphores_activity.py share_memory.py .gitignore U semaphore_sok_kimheng.md X
semaphore_sok_kimheng.md > abc ### ID: e20221601
1 ### Name: SOK KIMHENG
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS PLAYWRIGHT Python + v [ ] [ ] ... | [ ] [ ]
PS C:\Users\MSI\Documents\I4-GIC2025\Operating System> & C:/Users/MSI/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/MSI/Documents/I4-GIC2025/Operating System/share_memory.py"
Produced pair 0: P1-0, P2-0
Consumed pair: P2-0, P1-0
Produced pair 1: P1-1, P2-1
Consumed pair: P2-1, P1-1
Produced pair 2: P1-2, P2-2
Consumed pair: P2-2, P1-2
Produced pair 3: P1-3, P2-3
Consumed pair: P2-3, P1-3
Produced pair 4: P1-4, P2-4
Consumed pair: P2-4, P1-4
Produced pair 5: P1-5, P2-5
Consumed pair: P2-5, P1-5
Produced pair 6: P1-6, P2-6
Consumed pair: P2-6, P1-6
Produced pair 7: P1-7, P2-7
Consumed pair: P2-7, P1-7
Produced pair 8: P1-8, P2-8
Consumed pair: P2-8, P1-8
Produced pair 9: P1-9, P2-9
Consumed pair: P2-9, P1-9
Produced pair 10: P1-10, P2-10
Consumed pair: P2-10, P1-10
Produced pair 11: P1-11, P2-11
Consumed pair: P2-11, P1-11
```

- Problem 2

Result



The image shows a Visual Studio Code editor window with a Python file named `semaphores_activity.py` open. The file contains three functions: `process1()`, `process2()`, and `process3()`. `process1()` acquires semaphore `a`, prints 'H' and 'E', releases `a`, releases semaphore `b`, and sleeps for 1 second. `process2()` acquires semaphore `b`, prints 'L', releases `b`, releases semaphore `c`, and sleeps for 1 second. `process3()` acquires semaphore `c` twice, prints 'O', releases `c`, releases semaphore `a`, and sleeps for 1 second. The terminal at the bottom shows the command to run the script, and the output displays the characters 'H', 'E', 'L', 'L', 'O' on separate lines, followed by a blank line.

```
semaphores_activity.py > process3
6  def process1():
7      while True:
8          a.acquire()
9          print('H')
10         print('E')
11         b.release()
12         b.release()
13         time.sleep(1)
14
15
16  def process2():
17      global b, c
18      while True:
19          b.acquire()
20          print('L')
21          c.release()
22          time.sleep(1)
23
24  def process3():
25      global c, a
26      while True:
27          c.acquire()
28          c.acquire()
29          print('O')
30          print(' ')
31          a.release()
32          time.sleep(1)
33
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS PLAYWRIGHT Python + - [] [X] [] [X]

PS C:\Users\MSI\Documents\I4-GIC2025\Operating System> & C:/Users/MSI/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/MSI/Documents/I4-GIC2025/Operating System/semaphores_activity.py"

H
E
L
L
O

[]